Practice Examination 1 – Answers

Total 60 marks. Award method marks shown in brackets; an answer without working earns the final mark only.

Section A – Equations			
Q1 One-step equations	a	**x = 13**	x = 21 − 8
	b	**m = 17**	m = 5 + 12
	c	**p = 8**	p = 56 ÷ 7
	d	**y = 36**	y = 9 × 4
Q2 Two-step equations	a	**x = 7**	5x = 35 (1 mark), x = 35 ÷ 5 = 7 (1 mark)
	b	**x = 30**	x ÷ 6 = 5 (1 mark), x = 5 × 6 = 30 (1 mark)
Q3 Negative solution		**m = −4**	3m = 2 − 14 = −12 (1 mark), m = −12 ÷ 3 = −4 (1 mark)
Q4 Checking solutions		**NO**	4 × 6 − 9 = 24 − 9 = 15 (1 mark). 15 ≠ 13, so x = 6 is not the solution (1 mark)
Q5 Worded problem	a	**3n + 8 = 35**	
	b	**n = 9**	3n = 27 (1 mark), n = 9 (1 mark)
Q6 Worded problem	a	**4t + 7 = 55**	
	b	**t = 12, so one ticket costs $12**	4t = 48 (1 mark), t = 12 (1 mark)
Section B – Investigating Data			
Q7 Column graph	a	**Mango**	
	b	**46 students**	11 + 7 + 14 + 9 + 5 = 46
Q8 Line graph	a	**2 pm**	
	b	**9°C**	15 − 6 = 9
Q9 Sector graph	a	**60 students**	30% of 200 = 0.3 × 200 = 60
	b	**50 students**	Maths 35% of 200 = 70, Art 10% of 200 = 20 (1 mark), 70 − 20 = 50 (1 mark)
Q10 Stem-and-leaf plot	a	**27**	13 values, so the median is the 7th value
	b	**6**	values above 30: 31, 34, 36, 39, 42, 45
Q11 Dot plot	a	**1**	The tallest column of dots
	b	**6**	Count the dots at 3 and above
Q12 Frequency table	a	**38**	3 + 6 + 8 + 11 + 7 + 3 = 38
	b	**3**	3 has the highest frequency (11) – the answer is the value, not the frequency
Q13 Mean, median, mode, range	a	**11**	Total = 88 (1 mark), 88 ÷ 8 = 11 (1 mark)
	b	**10**	In order: 6, 8, 9, 9, 11, 13, 15, 17. Middle two values: 9 and 11
	c	**9**	Appears most often
	d	**11**	17 − 6
Q14 Outliers	a	**34**	Much further from the other values than any other score
	b	**11**	66 ÷ 6
	c	**Higher**	The outlier is much larger than the rest, so it pulls the mean up. Without it the mean is about 6.4.
Section C – Mixed Topics			
Q15 Pythagoras' theorem	a	**x = 15 cm**	$x^2 = 9^2 + 12^2 = 81 + 144 = 225$ (1 mark), $x = \sqrt{225} = 15$ (1 mark)
	b	**YES**	$10^2 + 24^2 = 100 + 576 = 676$ and $26^2 = 676$ (1 mark). They are equal, so it is right-angled (1 mark)
Q16 Decimals	a	**0.9**	36 × 25 = 900, 3 decimal places → 0.900
	b	**60**	5.4 ÷ 0.09 = 540 ÷ 9 = 60

Q17 Powers and roots	a	**125**	5 × 5 × 5
	b	**11**	11 × 11 = 121
Q18 Surds	a	**√18 and √7**	16 and 36 are square numbers, so √16 = 4 and √36 = 6 are not surds
	b	**4√3**	√48 = √16 × √3 (1 mark) = 4√3 (1 mark)
Q19 Algebra	a	**2x + 9y**	6x − 4x = 2x, 2y + 7y = 9y
	b	**5a + 15**	
	c	**63**	$3 \times 5^2 - 2 \times 6 = 3 \times 25 - 12 = 75 - 12$
Q20 Area and volume	a	**56 cm²**	½ × (6 + 10) × 7 (1 mark) = ½ × 16 × 7 = 56 (1 mark)
	b	**80 cm³**	8 × 5 × 2
Q21 Linear relationships		**y = −5, −2, 1, 4**	e.g. x = −2: 3 × (−2) + 1 = −6 + 1 = −5. 2 marks all correct, 1 mark for 2–3 correct
Q22 Rates and ratios	a	**3 : 5**	Divide both by 5 (the HCF)
	b	**18 km/h**	72 ÷ 4 = 18

Practice Examination 2 – Answers

Total 60 marks. Award method marks shown in brackets; an answer without working earns the final mark only.

Section A – Equations			
Q1 One-step equations	a	**x = 17**	x = 30 − 13
	b	**m = 27**	m = 19 + 8
	c	**p = 7**	p = 63 ÷ 9
	d	**y = 48**	y = 8 × 6
Q2 Two-step equations	a	**x = 6**	6x = 36 (1 mark), x = 6 (1 mark)
	b	**x = 20**	x ÷ 5 = 4 (1 mark), x = 4 × 5 = 20 (1 mark)
Q3 Negative solution		**m = −5**	4m = 5 − 25 = −20 (1 mark), m = −20 ÷ 4 = −5 (1 mark)
Q4 Checking solutions		**YES**	2 × 5 + 11 = 10 + 11 = 21 (1 mark). 21 = 21, so x = 5 is the solution (1 mark)
Q5 Worded problem	a	**5n − 9 = 41**	
	b	**n = 10**	5n = 50 (1 mark), n = 10 (1 mark)
Q6 Worded problem	a	**6h + 8 = 38**	
	b	**h = 5, so Ava hired the bike for 5 hours**	6h = 30 (1 mark), h = 5 (1 mark)
Section B – Investigating Data			
Q7 Column graph	a	**Thursday**	
	b	**100 books**	15 + 25 + 10 + 30 + 20 = 100
Q8 Line graph	a	**2 pm**	
	b	**35 people**	50 − 15 = 35
Q9 Sector graph	a	**56 people**	35% of 160 = 0.35 × 160 = 56
	b	**64 people**	Dog 45% of 160 = 72, Bird 5% of 160 = 8 (1 mark), 72 − 8 = 64 (1 mark)
Q10 Stem-and-leaf plot	a	**26**	13 values, so the median is the 7th value
	b	**4**	values above 30: 33, 37, 41, 46
Q11 Dot plot	a	**1**	The tallest column of dots
	b	**7**	Count the dots at 3 and above
Q12 Frequency table	a	**40**	12 + 15 + 6 + 4 + 3 = 40
	b	**2**	2 has the highest frequency (15) – the answer is the value, not the frequency
Q13 Mean, median, mode, range	a	**22**	Total = 132 (1 mark), 132 ÷ 6 = 22 (1 mark)
	b	**22.5**	In order: 15, 18, 21, 24, 24, 30. Middle two values: 21 and 24
	c	**24**	Appears most often
	d	**15**	30 − 15
Q14 Outliers	a	**4**	Much further from the other values than any other score
	b	**21**	126 ÷ 6
	c	**Lower**	The outlier is much smaller than the rest, so it pulls the mean down. Without it the mean is about 24.4.
Section C – Mixed Topics			
Q15 Pythagoras' theorem	a	**x = 20 cm**	$x^2 = 12^2 + 16^2 = 144 + 256 = 400$ (1 mark), $x = \sqrt{400} = 20$ (1 mark)
	b	**NO**	$6^2 + 9^2 = 36 + 81 = 117$ but $11^2 = 121$ (1 mark). $117 \neq 121$, so it is not right-angled (1 mark)

Q16 Decimals	a	**0.39**	26 × 15 = 390, 3 decimal places → 0.390
	b	**120**	9.6 ÷ 0.08 = 960 ÷ 8 = 120
Q17 Powers and roots	a	**32**	2 × 2 × 2 × 2 × 2
	b	**12**	12 × 12 = 144
Q18 Surds	a	**√20 and √30**	49 and 100 are square numbers, so √49 = 7 and √100 = 10 are not surds
	b	**6√2**	√72 = √36 × √2 (1 mark) = 6√2 (1 mark)
Q19 Algebra	a	**10m + 4n**	7m + 3m = 10m, −2n + 6n = 4n
	b	**4y − 20**	
	c	**20**	$3 \times 3^2 - 7 = 3 \times 9 - 7 = 27 - 7$
Q20 Area and volume	a	**66 cm²**	½ × (9 + 13) × 6 (1 mark) = ½ × 22 × 6 = 66 (1 mark)
	b	**180 cm³**	12 × 3 × 5
Q21 Linear relationships		**y = −9, −5, −1, 3**	e.g. x = −2: 4 × (−2) − 1 = −8 − 1 = −9. 2 marks all correct, 1 mark for 2–3 correct
Q22 Rates and ratios	a	**5 : 8**	Divide both by 4 (the HCF)
	b	**90 km/h**	450 ÷ 5 = 90

Practice Examination 3 – Answers

Total 60 marks. Award method marks shown in brackets; an answer without working earns the final mark only.

Section A – Equations			
Q1 One-step equations	a	**x = 14**	x = 33 − 19
	b	**m = 34**	m = 23 + 11
	c	**p = 9**	p = 72 ÷ 8
	d	**y = 33**	y = 11 × 3
Q2 Two-step equations	a	**x = 6**	7x = 42 (1 mark), x = 6 (1 mark)
	b	**x = 32**	x ÷ 4 = 8 (1 mark), x = 8 × 4 = 32 (1 mark)
Q3 Negative solution		**m = −6**	5m = 2 − 32 = −30 (1 mark), m = −30 ÷ 5 = −6 (1 mark)
Q4 Checking solutions		**NO**	5 × 7 − 8 = 35 − 8 = 27 (1 mark). 27 ≠ 26, so x = 7 is not the solution (1 mark)
Q5 Worded problem	a	**6n + 11 = 59**	
	b	**n = 8**	6n = 48 (1 mark), n = 8 (1 mark)
Q6 Worded problem	a	**9s + 65 = 290**	
	b	**s = 25, so 25 students went**	9s = 225 (1 mark), s = 25 (1 mark)
Section B – Investigating Data			
Q7 Column graph	a	**Pepperoni**	
	b	**41 students**	8 + 13 + 6 + 10 + 4 = 41
Q8 Line graph	a	**15 minutes**	
	b	**45°C**	90 − 45 = 45
Q9 Sector graph	a	**72 students**	30% of 240 = 0.3 × 240 = 72
	b	**96 students**	Pop 50% of 240 = 120, Rock 10% of 240 = 24 (1 mark), 120 − 24 = 96 (1 mark)
Q10 Stem-and-leaf plot	a	**26.5**	12 values, so the median is the average of the 6th and 7th values
	b	**6**	values above 25: 28, 30, 32, 34, 37, 43
Q11 Dot plot	a	**2**	The tallest column of dots
	b	**7**	Count the dots at 3 and above
Q12 Frequency table	a	**32**	7 + 13 + 8 + 3 + 1 = 32
	b	**1**	1 has the highest frequency (13) – the answer is the value, not the frequency
Q13 Mean, median, mode, range	a	**7**	Total = 70 (1 mark), 70 ÷ 10 = 7 (1 mark)
	b	**7.5**	In order: 3, 4, 5, 6, 7, 8, 8, 8, 10, 11. Middle two values: 7 and 8
	c	**8**	Appears most often
	d	**8**	11 − 3
Q14 Outliers	a	**460**	Much further from the other values than any other score
	b	**180**	1080 ÷ 6
	c	**Higher**	The outlier is much larger than the rest, so it pulls the mean up. Without it the mean is about 124.0.
Section C – Mixed Topics			
Q15 Pythagoras' theorem	a	**x = 10.8 cm**	$x^2 = 6^2 + 9^2 = 36 + 81 = 117$ (1 mark), $x = \sqrt{117} = 10.816\ldots \approx 10.8$ (1 mark)
	b	**YES**	$9^2 + 40^2 = 81 + 1600 = 1681$ and $41^2 = 1681$ (1 mark). They are equal, so it is right-angled (1 mark)

Q16 Decimals	a	**0.5**	125 × 4 = 500, 3 decimal places → 0.500
	b	**6**	4.2 ÷ 0.7 = 42 ÷ 7 = 6
Q17 Powers and roots	a	**1000**	10 × 10 × 10
	b	**15**	15 × 15 = 225
Q18 Surds	a	**√27 and √14**	81 and 64 are square numbers, so √81 = 9 and √64 = 8 are not surds
	b	**5√3**	√75 = √25 × √3 (1 mark) = 5√3 (1 mark)
Q19 Algebra	a	**5k + 8**	9k − 4k = 5k, 5 + 3 = 8
	b	**12x + 6**	
	c	**6**	$5 \times 2^2 - 2 \times 7 = 5 \times 4 - 14 = 20 - 14$
Q20 Area and volume	a	**36 cm²**	½ × (11 + 7) × 4 (1 mark) = ½ × 18 × 4 = 36 (1 mark)
	b	**360 cm³**	6 × 6 × 10
Q21 Linear relationships		**y = −8, −3, 2, 7**	e.g. x = −2: 5 × (−2) + 2 = −10 + 2 = −8. 2 marks all correct, 1 mark for 2–3 correct
Q22 Rates and ratios	a	**2 : 3**	Divide both by 14 (the HCF)
	b	**12 km/h**	30 ÷ 2.5 = 12

Practice Examination 4 – Answers

Total 60 marks. Award method marks shown in brackets; an answer without working earns the final mark only.

Section A – Equations			
Q1 One-step equations	a	**x = 26**	x = 50 − 24
	b	**m = 23**	m = 6 + 17
	c	**p = 7**	p = 84 ÷ 12
	d	**y = 40**	y = 5 × 8
Q2 Two-step equations	a	**x = 6**	8x = 48 (1 mark), x = 6 (1 mark)
	b	**x = 14**	x ÷ 2 = 7 (1 mark), x = 7 × 2 = 14 (1 mark)
Q3 Negative solution		**m = −7**	2m = 5 − 19 = −14 (1 mark), m = −14 ÷ 2 = −7 (1 mark)
Q4 Checking solutions		**YES**	6 × 3 + 4 = 18 + 4 = 22 (1 mark). 22 = 22, so x = 3 is the solution (1 mark)
Q5 Worded problem	a	**n ÷ 2 + 7 = 19**	also accept n/2 + 7 = 19
	b	**n = 24**	n ÷ 2 = 12 (1 mark), n = 24 (1 mark)
Q6 Worded problem	a	**25m + 40 = 290**	
	b	**m = 10, so Zara had the plan for 10 months**	25m = 250 (1 mark), m = 10 (1 mark)
Section B – Investigating Data			
Q7 Column graph	a	**March**	
	b	**45 cars**	9 + 6 + 12 + 7 + 11 = 45
Q8 Line graph	a	**4 am**	
	b	**10°C**	8 − (−2) = 8 + 2 = 10
Q9 Sector graph	a	**30 students**	10% of 300 = 30
	b	**90 students**	Gaming 35% of 300 = 105, Reading 5% of 300 = 15 (1 mark), 105 − 15 = 90 (1 mark)
Q10 Stem-and-leaf plot	a	**61**	13 values, so the median is the 7th value
	b	**5**	values of 64 or more: 64, 66, 69, 70, 74
Q11 Dot plot	a	**3**	The tallest column of dots
	b	**6**	Count the dots above 3 (not including 3)
Q12 Frequency table	a	**25**	5 + 8 + 6 + 4 + 2 = 25
	b	**1**	1 has the highest frequency (8) – the answer is the value, not the frequency
Q13 Mean, median, mode, range	a	**24**	Total = 192 (1 mark), 192 ÷ 8 = 24 (1 mark)
	b	**24**	In order: 19, 19, 19, 23, 25, 27, 29, 31. Middle two values: 23 and 25
	c	**19**	Appears most often
	d	**12**	31 − 19
Q14 Outliers	a	**8**	Much further from the other values than any other score
	b	**37**	222 ÷ 6
	c	**Lower**	The outlier is much smaller than the rest, so it pulls the mean down. Without it the mean is about 42.8.
Section C – Mixed Topics			
Q15 Pythagoras' theorem	a	**x = 34 cm**	$x^2 = 16^2 + 30^2 = 256 + 900 = 1156$ (1 mark), $x = \sqrt{1156} = 34$ (1 mark)
	b	**NO**	$5^2 + 11^2 = 25 + 121 = 146$ but $12^2 = 144$ (1 mark). $146 \neq 144$, so it is not right-angled (1 mark)

Q16 Decimals	a	**0.056**	7 × 8 = 56, 3 decimal places → 0.056
	b	**90**	8.1 ÷ 0.09 = 810 ÷ 9 = 90
Q17 Powers and roots	a	**216**	6 × 6 × 6
	b	**14**	14 × 14 = 196
Q18 Surds	a	**√45 and √8**	1 and 121 are square numbers, so √1 = 1 and √121 = 11 are not surds
	b	**4√5**	√80 = √16 × √5 (1 mark) = 4√5 (1 mark)
Q19 Algebra	a	**$2x^2 + 7x$**	$3x^2 - x^2 = 2x^2$, 5x + 2x = 7x (x^2 and x are NOT like terms)
	b	**−2x + 12**	−2 × x = −2x, −2 × (−6) = +12
	c	**31**	$4 \times (-2)^2 + 3 \times 5 = 4 \times 4 + 15 = 16 + 15$
Q20 Area and volume	a	**42 cm²**	½ × (5.5 + 8.5) × 6 (1 mark) = ½ × 14 × 6 = 42 (1 mark)
	b	**150 cm³**	15 × 4 × 2.5
Q21 Linear relationships		**y = 3, 1, −1, −3**	e.g. x = −1: −2 × (−1) + 1 = 2 + 1 = 3. 2 marks all correct, 1 mark for 2–3 correct
Q22 Rates and ratios	a	**3 : 4**	Divide both by 15 (the HCF)
	b	**78 km/h**	117 ÷ 1.5 = 78